

Bus Stop Location Optimization Tool

Welcome to the Bus Stop Location Optimization Tool. This interactive dashboard allows urban planners and decision-makers to design efficient public transport networks by balancing financial costs with public accessibility.

Follow these three steps to use the tool effectively:

1. Configure Your Inputs

The tool provides flexible input parameters so you can customize the optimization model to fit specific needs and budgets

1

Weighting Factor (Alpha)

Use the slider to balance the two competing objectives. A higher value prioritizes the Benefit for Society (maximizing coverage and minimizing walking distance), while a lower value prioritizes Profit/Cost Efficiency (minimizing the number of built stops).

2

Max Walking Distance

Set the maximum acceptable distance (in meters) a citizen should have to walk to reach a bus stop.

3

Max Bus Stops

Set a strict upper limit on how many bus stops can be built, acting as your budget constraint.

4

Minimum Coverage Percentage

Guarantee a baseline service level by forcing the model to cover at least this percentage of the total population.

5

Fixed Cost per Bus Stop

Input the financial cost of building a single stop to calculate total network investments.

Tip: Switch from MCLP to LSCP

By default, the tool solves a Maximal Covering Location Problem (MCLP) to maximize coverage within your budget. However, you can easily switch the tool to solve a Location Set Covering Problem (LSCP)—which finds the absolute minimum cost required to serve the entire city. To do this: Set the Minimum Coverage Percentage to 100% and the Weighting Factor (Alpha) to 0.

2. Analyze the Results

Once the model runs, you can evaluate the proposed network through two main views:

KPI Dashboard

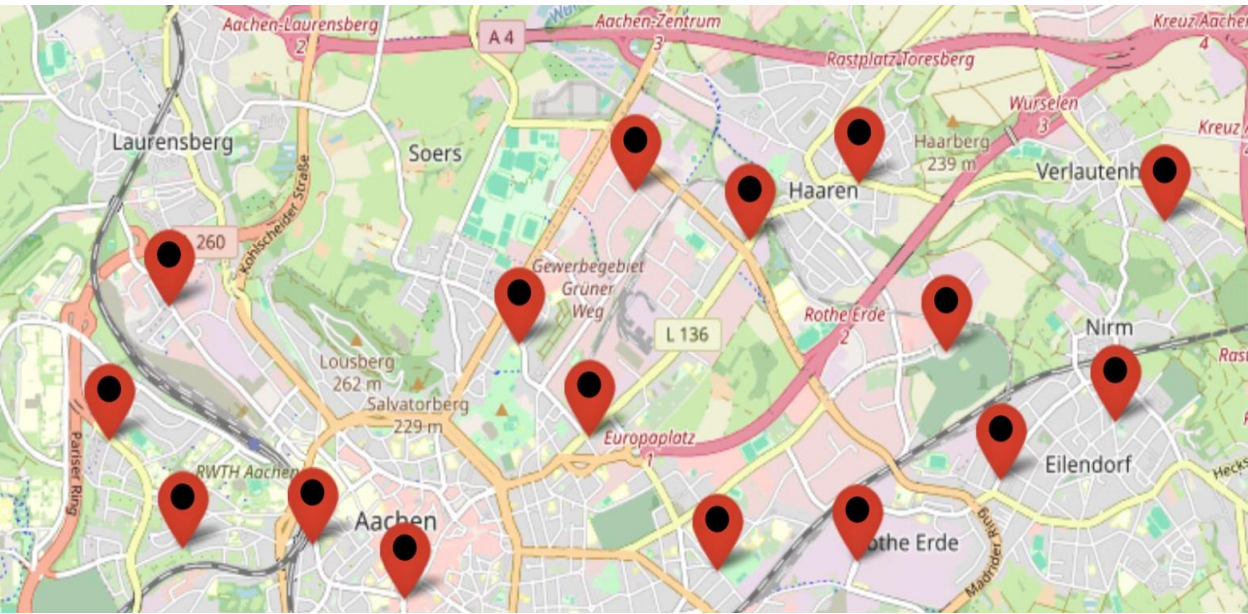
Review high-level metrics, including total population served, average walking distance, and total construction costs.



Interactive Maps

Visually inspect the spatial distribution of the network.

- Map 1: Visualizes all potential candidate locations for bus stops.
- Map 2: Shows only the optimal bus stops that the model selected to be built, allowing you to see exactly where investments will go.



3. Compare Scenarios

Decision-making requires understanding trade-offs. The tool allows you to save and compare different optimization runs side-by-side.

Evaluate the marginal change in each objective (e.g., "How much more coverage do we get if we build exactly one more stop?").

Compare a heavily budget-constrained scenario against a highly service-oriented scenario to find the perfect political and financial middle ground.

